WORKSHEET - NAPLAN Year 9

Number and Algebra
Solving Equations (9)

Teacher: Joanne Rust
Exam Equivalent Time: 10 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Number and Algebra, NAP-55614

Zilda has 4 children.

Each child needs 6 pens when they start school for the new year.

Zilda has no pens when she goes to the shop.

The shop sells its pens in packets of 5.

How many packets does Zilda need to buy?

2 4 5 6

2. Number and Algebra, NAP-49698

Olivia earned $\$17.24$ per hour working at a pizza store.

This week she worked for $8\frac{1}{4}$ hours.

She used the money she earned this week to buy concert tickets for herself and her friends.

Each concert ticket cost $\$14.15$

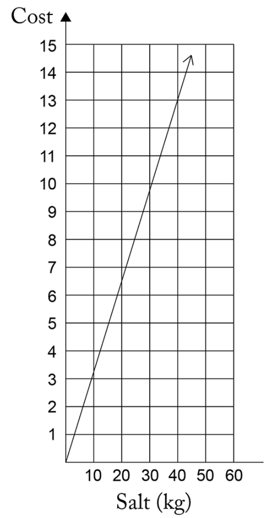
What is the maximum number of concert tickets Olivia could have bought?

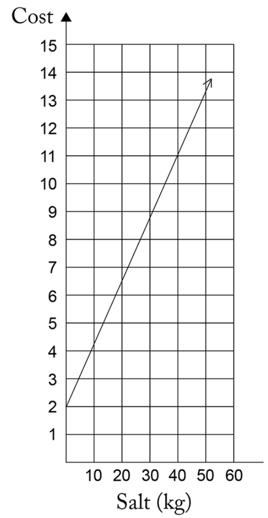
[ ] tickets

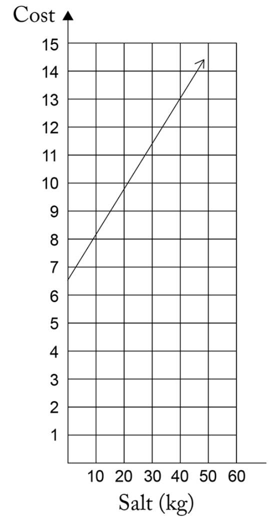
3. Number and Algebra, NAP-43786

At a pool supply store, 20 kilograms of salt costs \$6.50 and 40 kilograms of the same salt costs \$13.00.

Which graph could show the relationship between the amount of salt and its cost?







4. Number and Algebra, NAP-55644

Lachlan drives his boat to an island 100 km away at an average speed of 80 km/h.

How long does the trip take?

48 minutes 1 hour 15 minutes 1 hour 20 minutes 1 hour 30 minutes

5. Number and Algebra, NAP-02686

Sophia and Cailen sold raffle tickets to fund their trip to the National Swimming Titles.

If Cailen had collected $20 more, he would have collected exactly twice as much as Sophia.

Which row of the table shows how much money they could have collected?

Sophia	Cailen
$20	$40
$25	$40
$30	$40
$40	$70

6. Number and Algebra, NAP-79169

A plumber calculates the price of a job using a service fee and an amount per hour.

This table shows some of the job prices.

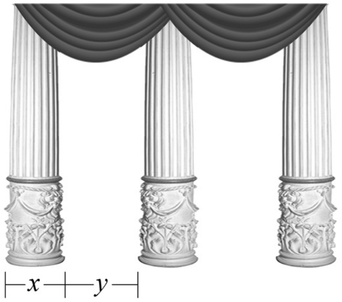
Hours	2	4	6	8
Job price	$130	$202	$274	$346

How are the jobs calculated?

- $60 service fee + $35 per hour
- $60 service fee + $70 per hour
- $58 service fee + $72 per hour
- $58 service fee + $36 per hour

7. Number and Algebra, NAP-02746

A reception area has pillars that are x cm wide and the gap between pillars is y cm.



If the reception area has 12 pillars on one side, the length of that side can be represented by which expression?

- $11(x+y)$
- $12(x+y)$
- $12x+y$
- $12xy$
- $12x+11y$

8. Number and Algebra, NAP-55733

Michelle creates a snack food which is a mixture of 4 different types of nuts.

She makes 300 grams of the mixture and weighs each ingredient as follows:

- 40% of the mixture is almonds.
- The rest of the mixture is made up of walnuts, pecans and macadamia nuts.
- The mass of walnuts is the same as the mass of pecans.
- The mass of walnuts is twice the mass of macadamias.

How many grams of pecans are in the 300 grams of nut mixture.

_______ grams

Worked Solutions

1. Number and Algebra, NAP-55614

$$\begin{aligned}\text{Pens required} &= 4 \times 6\\ &= 24\end{aligned}$$

$$\begin{aligned}\therefore\ \text{Packets required} &= \frac{24}{5}\\ &= 4.8\\ &= 5\end{aligned}$$

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2. Number and Algebra, NAP-49698

$$\begin{aligned}\text{Money earned} &= 8\frac{1}{4} \times 17.24\\ &= 142.23\end{aligned}$$

$$\begin{aligned}\therefore\ \text{Maximum tickets} &= \frac{142.23}{14.15}\\ &= 10.05\ldots\\ &= 10\end{aligned}$$

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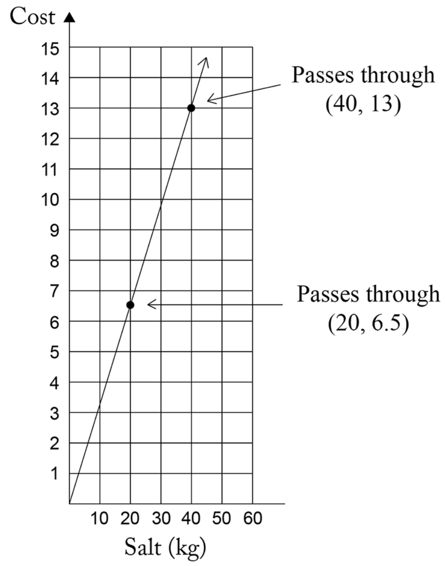
3. Number and Algebra, NAP-43786



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4. Number and Algebra, NAP-55644

$$\begin{aligned}\text{Time} &= \frac{\text{Distance}}{\text{Speed}}\\ &= \frac{100}{80}\\ &= 1.25 \text{ hours}\\ &= 1 \text{ hour } 15 \text{ minutes}\end{aligned}$$

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5. Number and Algebra, NAP-02686

$$\begin{aligned}\text{Cailen } + \$20 &= 2 \times \text{Sophia}\\ \$40 + \$20 &= 2 \times \$30\\ \$60 &= \$60\end{aligned}$$

$\therefore$ Sophia \$30, Callen \$40

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6. Number and Algebra, NAP-79169

Testing Option 4 equation with the table values:

$130 = 58 + 2 \times 36$

$202 = 58 + 4 \times 36$

$274 = 58 + 6 \times 36$

$346 = 58 + 8 \times 36$

$\therefore$ \$58 service fee $+$ \$36 per hour

7. Number and Algebra, NAP-02746

The length of the side

$= $ width of 12 pillars $+$ width of 11 gaps

$= 12x + 11y$

8. Number and Algebra, NAP-55733

Weight of almonds $= 40\% \times 300$

$= 120$ grams

Weight of remaining nut mixture

$= 300 - 120$

$= 180$ g

Let $x =$ weight of pecans

$$x + x + \frac{1}{2}x = 180$$

$$\therefore x = \frac{180}{2.5}$$

$$= 72 \text{ grams}$$